📩 Writing Effective User Instructions

Clear and Specific Instructions

Principle

Ambiguous prompts produce ambiguous outputs.

The model performs best when:

- The task is explicit
- The expected output format is defined
- Constraints are clearly stated

Example

❌ **Vague Prompt:**
"Explain AI."

✅ **Precise Prompt:**
"Explain how neural networks work in 5 bullet points for beginners."

🔹 Delimiters

Definition

Delimiters are characters used to separate instructions from data, preventing confusion.

They help the model distinguish:

- What to do
- What to analyze

Common Delimiters

===
'''
###
***
<data></data>

Example Usage

Summarize the text below in 3 bullet points.

###
Artificial intelligence is transforming industries by automating decision-making...
###

This ensures the model clearly identifies the input boundary.

🎯 The X–Y Problem

Definition

The X–Y problem occurs when someone asks about a solution (Y) instead of the actual goal (X).

Structure

- **X = The real objective**
- **Y = The attempted solution**

Example

❌ “How do I remove spaces from a string?”

✔ **Real goal:**
“How do I parse structured data efficiently?”

Key Rule

Always communicate goals, not just attempted solutions.

This improves response relevance significantly.

🧠 In-Context Learning

Definition

In-context learning is the model’s ability to learn patterns from examples within the prompt itself, without retraining.

Types of In-Context Prompting

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1. Zero-Shot Prompting

Definition

No examples provided — only instructions.

Example

Translate this sentence to French:
"Hello, how are you?"

2. One-Shot Prompting

Definition

One example is provided to guide the model.

Example

English: Cat
French: Chat

English: Dog
French:

3. Few-Shot Prompting

(4–8 examples are typically sufficient for strong performance)

Definition

Multiple examples are provided to establish a pattern.

Example

2 + 3 = 5
4 + 6 = 10
7 + 8 =

Why It Works

Examples help the model:

- Understand task structure
- Reduce ambiguity

- Improve accuracy

Important Insight

Even if examples are incorrect, the model often follows the pattern they establish.

Examples primarily signal **how the output should look**, not whether the logic is correct.

🔗 Chain-of-Thought Prompting (COT)

Definition

Chain-of-Thought prompting involves asking the model to show the reasoning steps it uses to reach an answer.

Key Benefits

- Produces more accurate answers
- Reveals how the model approaches a problem
- Enables debugging of reasoning processes

Usage Insight

The user can provide reasoning steps as examples to guide the model.

If using **zero-shot prompting**, simply instructing:

> “Think step by step before answering.”

is often sufficient to improve results.